

Workshop On Deep Learning - QQQ Forecasting with Chronos-2

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Abstract

We study whether market, macroeconomic, calendar, disclosure, and news signals improve probabilistic forecasts of QQQ daily log returns. The forecasting model is Amazon Chronos-2, evaluated both with its original pretrained weights and after parameter-efficient LoRA fine-tuning. All full experiments use 2,133 training sessions and predict five separate daily returns at each of 50 non-overlapping validation origins, giving 250 outcomes. Across three seeds, the model using all external sources achieved mean weighted quantile loss 0.689439 versus 0.693582 for the control. In the single-topic GDELT screen, Federal Reserve news ranked first at 0.686526, narrowly ahead of recession news at 0.686796. The differences between the leading topics are not yet statistically resolved, so the results support further testing rather than a trading claim.

1 Introduction

Daily equity returns are noisy, weakly predictable, and dominated by changing volatility regimes. These properties create a difficult test for time-series foundation models. A useful forecast must describe uncertainty as well as a central estimate; a model that predicts a small return near zero can look competitive under point-error metrics while contributing little information about risk.

Our project asks whether external information tells a pretrained forecasting model how the distribution of future Nasdaq-100 returns should change. The target is the one-day adjusted-close log return of QQQ. At every origin, the model directly predicts distributions for trading days one through five. This is a five-output forecast, not a cumulative weekly-return prediction.

The original proposal considered Bitcoin and a Temporal Fusion Transformer. We changed the asset to QQQ because public macro, Federal Reserve, SEC, volatility, and technology-news data provide a stronger setting for controlled covariate experiments. Chronos-2 replaced the custom architecture because it supplies a pretrained baseline, supports past and known-future covariates, and can be adapted with a small LoRA checkpoint [1, 2].

The main contribution is an executed ablation study. Every news setup starts from the same pinned Chronos-2 revision, keeps a common 21-feature market and calendar control, trains only on data before the validation interval, and is scored on the same 250 realized returns. This design lets us ask which source or narrative adds information beyond observable market conditions.

2 Related work

2.1 Time-series foundation models

Chronos-2 is a pretrained universal forecasting model that accepts univariate series, multivariate groups, and covariates. Its group-attention mechanism shares information among the target and related series, enabling zero-shot covariate-informed forecasting [1]. We use the original pretrained model as a baseline and keep its revision fixed across every experiment.

Finance is a particularly strict application. Rahimikia, Ni, and Wang report that off-the-shelf time-series foundation models struggle on daily excess returns and that domain adaptation, dataset scale, and tuning materially affect performance [3]. This motivates our emphasis on calibration, matched controls, and repeated seeds instead of interpreting small point-estimate differences as dependable return predictability.

2.2 Low-rank adaptation

LoRA freezes the pretrained parameters and learns low-rank updates inside selected transformer layers [2]. It reduces the number of trainable parameters and allows one compact checkpoint per feature configuration. In this project, every setup receives a fresh LoRA adapter trained from the original base weights. Topic models never inherit the previously fine-tuned control adapter.

2.3 News and policy signals

News volume and tone can represent attention and sentiment, but they also contain overlapping and noisy signals. GDELT's DOC API reports topic coverage as a share of monitored global news and provides a tone timeline [5]. We retain both daily share and average tone for six topics: artificial intelligence, semiconductors, the Federal Reserve, inflation, big-tech earnings, and recession.

Economic Policy Uncertainty and Equity Market Uncertainty provide longer-running daily measures of news-based uncertainty [6]. Federal Reserve communication is represented separately by a sentence classifier that scores 170 policy statements as hawkish, dovish, or neutral [7]. SEC filing and earnings-event activity supplies a disclosure channel that differs from media coverage.

2.4 Evaluation principle

The primary score is weighted quantile loss (WQL), computed over 21 forecast quantiles and normalized by the mean absolute realized return. Lower WQL rewards a distribution that is both sharp and calibrated. We report mean absolute error (MAE), root mean squared error, directional accuracy, 10%-90% and 1%-99% interval coverage, and paired bootstrap intervals as supporting diagnostics.

$WQL = 2 \times \text{mean pinball loss} / \text{mean absolute realized return}$. Directional accuracy uses the median forecast only; it therefore answers a different question from WQL.

3 Dataset and data collection

The processed table contains 5,201 NYSE sessions from January 2006 through September 4, 2026, with 2005 QQQ history used to initialize rolling features. The controlled news experiments use the common interval January 3, 2017 through June 30, 2026 because all six GDELT topics are available through that date. The target is QQQ adjusted-close one-day log return.

3.1 Sources and roles

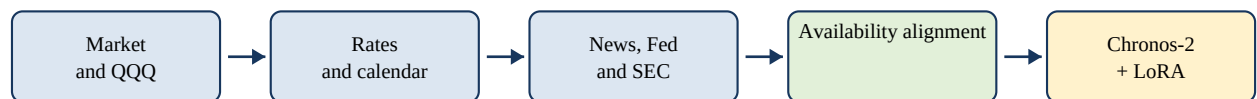
Source	Signals used	Role at forecast time
QQQ / Yahoo	Gap, range volatility, momentum, volume z-score, distance to 200-day mean	Past only
CBOE / ETFs	VXN change, VXN-VIX spread, VIX term ratio, TLT and HYG returns	Past only
FRED / ALFRED	2-year and real-yield changes, 10y-2y slope, vintage CPI year-over-year	Past only
Calendar	FOMC, CPI, NFP, month-end, options-expiration timing	Known future
GDELT	Daily news share and average tone for six technology and macro topics	Past only, T-1
Policy / filings	EPU, EMU, FOMC classifier state, earnings and SEC filing counts	Past only

Table 1: Main data sources and their role in the forecasting contract.

3.2 Point-in-time alignment

Each observation is assigned by the first QQQ close at which it could be known. Rates follow their publication schedule, CPI uses ALFRED release vintages, SEC forms use UTC acceptance timestamps, and completed daily news is lagged to the next market session. CBOE indices and daily commodity or currency bars receive a conservative one-day lag. Scheduled calendar fields are the only values allowed in the future input horizon.

Missing past data remains missing and reaches Chronos-2 through its observation mask. A failed news request is never converted into a zero count. This matters because a zero means that the source was observed and contained no matching news, whereas a missing value means the source was unavailable.



Five separate daily QQQ return distributions at each forecast origin

Fig. 1: Data flow from source-specific observations to the five-day probabilistic forecast.

3.3 Feature profiles

The complete table contains 55 past covariates and 14 known-future calendar values. The common control uses 21 selected signals: six QQQ transforms, five market variables, four rate and macro variables, and six calendar variables. A single-topic GDELT run adds exactly two columns - that topic's news share and tone - for 23 total features. The all-external setup contains 42 features.

4 Methodology

4.1 Forecasting protocol

Every full run uses the pinned *amazon/chronos-2* revision 29ec3766d36d6f73f0696f85560a422f50e8498c. At a validation origin, the model receives observations through the previous QQQ close and produces 21 quantiles for each of the next five trading-day returns. After those five realized sessions, the origin advances by five sessions. Fifty non-overlapping origins therefore score 250 daily outcomes; the final two sessions of the 252-session allocation cannot form a complete block.

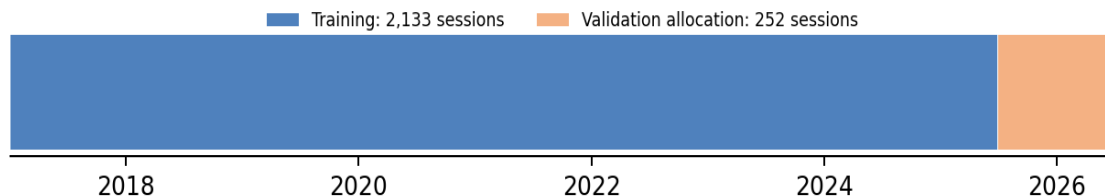


Fig. 2: Chronological train and development-validation split used by every full run.

4.2 Pretrained and fine-tuned arms

For each feature configuration we first evaluate the original pretrained weights on all 250 outcomes. We then restart from those same weights, train a fresh LoRA adapter on the 2,133 pre-validation sessions, and repeat the identical evaluation. Training uses context length 512, 200 optimizer steps, learning rate 1e-5, batch size 8, and seed 42 unless a robustness run explicitly selects seed 43 or 44. No validation outcome is used for checkpoint selection.

4.3 Experiment ladder

Stage	Executed configurations	Purpose
GPU smoke	Pretrained, five-step LoRA, Gaussian; 15 outcomes	Verify CUDA, training, save/reload and prediction
Source screen	Control; EPU/EMU; FOMC tone; SEC; all GDELT; all external	Measure source-family value at seed 42
Robustness	Control, EPU/EMU and all external at seeds 43 and 44	Check whether source ranking repeats
Topic screen	Six GDELT topics, one pair of share/tone columns per run	Identify the most informative narrative

Table 2: Executed experiment stages. The full stages comprise 18 LoRA fits.

4.4 Baselines and uncertainty

A zero-mean Gaussian baseline estimates volatility from the last 60 target returns at each origin. Paired comparisons preserve the 50 common forecast blocks and bootstrap their WQL differences. A 95% interval entirely below zero favors the candidate over control. Reusing the same development period for several comparisons still creates selection risk, so a later untouched block is required for a final claim.

5 Executed pretrained and fine-tuning results

5.1 Infrastructure smoke test

TAU Slurm job 869927 completed on an NVIDIA GeForce RTX 2080 Ti. On its deliberately small 15-outcome sample, WQL decreased from 0.734899 with pretrained weights to 0.710572 after five LoRA steps, a 3.31% improvement. MAE improved 1.84% and RMSE improved 2.08%. The Gaussian baseline remained best at WQL 0.667505. This run establishes execution correctness, not forecasting skill.

5.2 Seed-42 source-family screen

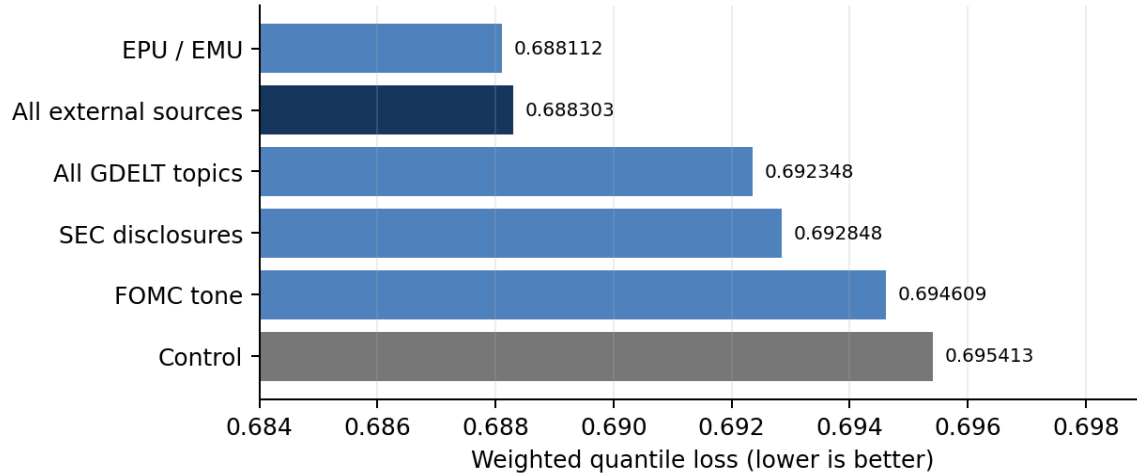


Fig. 3: Fine-tuned seed-42 WQL for the common control and five source configurations.

Setup	Features	WQL	Delta vs control	Direction	80% coverage
EPU / EMU	23	0.688112	-0.007301	57.2%	77.6%
All external	42	0.688303	-0.007109	58.0%	79.6%
All GDELT	33	0.692348	-0.003065	55.6%	77.6%
SEC disclosures	24	0.692848	-0.002565	58.0%	79.2%
FOMC tone	25	0.694609	-0.000804	57.6%	77.6%
Control	21	0.695413	0.000000	56.0%	78.0%

Table 3: Seed-42 source-family results. Lower WQL is better.

EPU/EMU ranked first at seed 42 and beat control in 64% of forecast blocks. Its paired interval was below zero. This initial result did not repeat at seeds 43 and 44, motivating the explicit robustness stage.

6 Robustness and GDELT topic attribution

6.1 Three-seed robustness

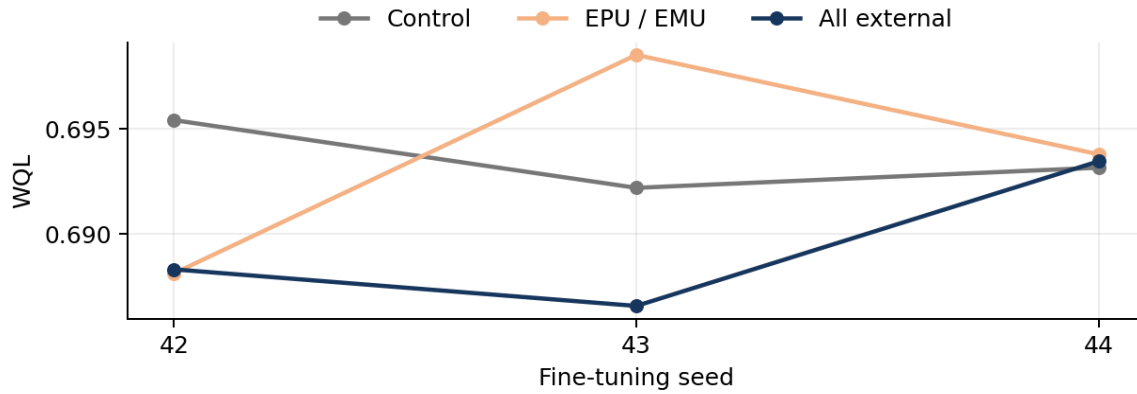


Fig. 4: WQL across fine-tuning seeds 42, 43, and 44 for the repeated configurations.

Across three seeds, all external sources produced mean WQL 0.689439 versus 0.693582 for control, a 0.60% improvement, and won two of three seeds. Its mean directional accuracy was 57.47% versus 56.67% for control. The aggregate paired interval $(-0.010021, 0.001424)$ crossed zero. EPU/EMU's strong seed-42 advantage disappeared in the repetitions, with mean WQL 0.693473.

6.2 Single-topic GDELT screen

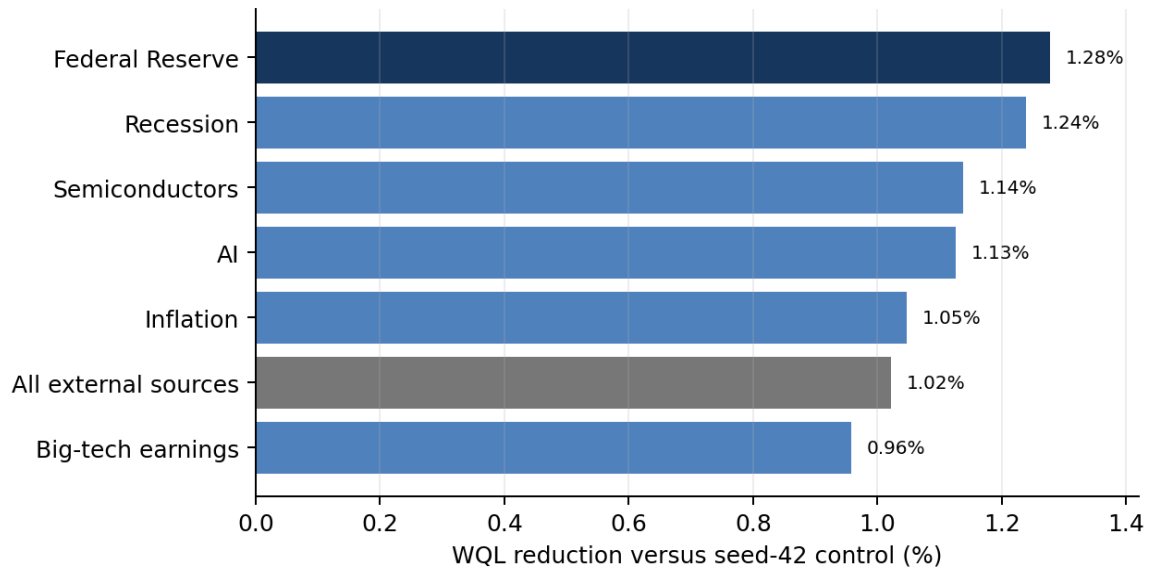


Fig. 5: WQL reduction versus the matched seed-42 control. The grey bar is the previous all-external model.

Rank	News data	WQL	Reduction	Correct direction
1	Federal Reserve	0.686526	1.28%	144 / 250
2	Recession	0.686796	1.24%	144 / 250
3	Semiconductors	0.687499	1.14%	144 / 250
4	AI	0.687580	1.13%	143 / 250
5	Inflation	0.688131	1.05%	142 / 250
6	All external sources	0.688303	1.02%	145 / 250
7	Big-tech earnings	0.688750	0.96%	142 / 250
-	Control: no news	0.695413	-	140 / 250

Table 4: Eight-case topic comparison on the same 250 daily outcomes.

7 Interpretation

Federal Reserve news has the best seed-42 WQL point estimate, with recession news nearly tied. All six isolated topics beat the fine-tuned control and their paired intervals against control were below zero. Direct comparisons among the topics remained unresolved: the Fed-minus-recession difference was -0.000269 with a 95% interval from -0.002163 to 0.001596 .

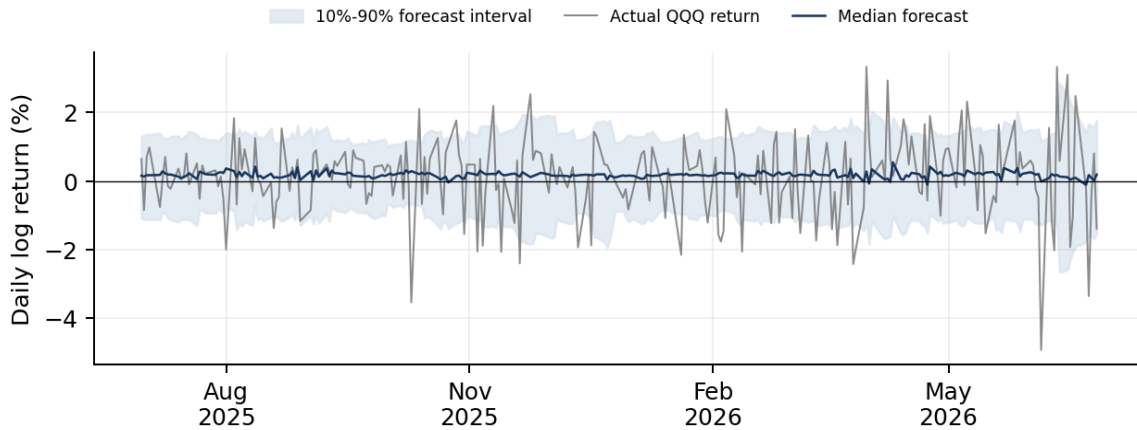


Fig. 6: Fed-topic fine-tuned forecasts across the 250 development outcomes.

The all-external model was not the worst case. It ranked sixth by WQL, ahead of big-tech earnings and control, and achieved the highest direction count at 145 of 250. Its distribution was wider than the Fed model and its improvement over its own pretrained arm was smaller. This pattern is consistent with redundant or noisy covariates diluting a cleaner signal, while still helping the sign of the median forecast.

7.1 Limitations

- The 2025-2026 interval is development validation and was reused for source selection.
- The topic screen has one fine-tuning seed; Fed and recession require repetitions.
- GDELT begins in 2017 and contains a June 2025 outage; unavailable observations remain masked.
- GDELT share and generic tone measure media coverage, not verified market-relevant sentiment.
- Economic, price, and filing histories can be revised; the pipeline applies conservative availability rules but is not a complete vintage archive.
- The SEC issuer basket uses current technology leaders and therefore contains survivorship hindsight.
- Chronos-2 pretraining may include historical market patterns; overlap cannot be ruled out.
- No transaction costs, portfolio construction, or tradability claim is evaluated.

7.2 Next experiments

The immediate next run should repeat Fed and recession at seeds 43 and 44, followed by a small Fed-plus-recession combination. The selected configuration should then be frozen and evaluated once on newly arriving sessions after September 4, 2026. Later work can test SPY and XLK as related targets, compare a quantile tree baseline, and obtain a more reliable point-in-time news archive.

8 Conclusion

The project establishes a reproducible Chronos-2 forecasting pipeline and executes 18 full LoRA fits plus a GPU smoke test. External information produced small improvements in probabilistic QQQ forecasts, and the topic screen points to Federal Reserve and recession coverage as the most promising narratives. The evidence is exploratory: the leading topics remain statistically tied and require repeated seeds and an untouched future evaluation before a final forecasting claim.

References

- [1] Ansari, A. F., et al.: *Chronos-2: From Univariate to Universal Forecasting*. arXiv:2510.15821 (2025). arxiv.org/abs/2510.15821
- [2] Hu, E. J., et al.: *LoRA: Low-Rank Adaptation of Large Language Models*. arXiv:2106.09685 (2021). arxiv.org/abs/2106.09685
- [3] Rahimikia, E., Ni, H., Wang, W.: *Re(Visiting) Time Series Foundation Models in Finance*. arXiv:2511.18578 (2025). arxiv.org/abs/2511.18578
- [4] Amazon Science: *Chronos Forecasting*, source code and model documentation. github.com/amazon-science/chronos-forecasting
- [5] The GDELT Project: *GDELT DOC 2.0 API*. blog.gdeltproject.org/gdelt-doc-2-0-api-debuts
- [6] Baker, S. R., Bloom, N., Davis, S. J.: Measuring Economic Policy Uncertainty. *Quarterly Journal of Economics* 131(4), 1593-1636 (2016).
- [7] Shah, A., Paturi, A., Chava, S.: Trillion Dollar Words: A New Financial Dataset, Task and Market Analysis for Central Bank Communications. *ACL* (2023).

Appendix A Executed run ledger

Slurm job	Runs	Configuration	Outcome
869927	1 smoke	5 LoRA steps; 15 outcomes	Completed; adapter reload and holdout prediction passed
869989	6	Source screen, seed 42	Completed; 250 outcomes per run
871482	6	Control, EPU/EMU, all external; seeds 43-44	Completed; three-seed robustness assembled
874192	6	One GDELT topic per run, seed 42	Completed; Fed ranked first

Table 5: Successful TAU Slurm executions included in this report.

Appendix B Reproducibility record

The full experiments use feature-table SHA-256 e0c8d48b38059eb70d9f91931bff68678dbcc46aeadaab1502ded1b9ee11df037. The verified cluster runtime was Python 3.12.3, PyTorch 2.11.0+cu128, CUDA 12.8, Chronos Forecasting 2.3.2, Transformers 5.16.1, PEFT 0.20.0, and Accelerate 1.14.0 on an RTX 2080 Ti. Each run directory stores its environment, data-readiness record, metadata, 250 predictions, per-horizon metrics, and LoRA adapter.

Primary local artifacts: `models/news-ablation-869989/`. Experiment definitions: `configs/news_ablation.json`. Readable analysis: `docs/NEWS_ABLATION_RESULTS.md` and `docs/GDELT_TOPIC_RESULTS.md`.